

Rihua Van Steenburgh

Data Engineer | AI & Analytics

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Summary

Data Engineer with hands-on experience building cloud data pipelines, analytics-ready datasets, and AI/RAG-enabled data applications using Python, SQL, Azure, AWS, Databricks, PySpark, dbt, PostgreSQL/pgvector, and Power BI. Strong in API ingestion, ETL/ELT, data validation, dimensional modeling, CI/CD, and technical documentation.

Skills

- **Languages:** Python, SQL, R, JavaScript, TypeScript
- **Cloud & Data Platforms:** AWS, Azure Data Factory, ADLS Gen2, Databricks, AWS S3, Redshift Serverless, ECS/Fargate, EventBridge, CloudWatch
- **Data Engineering:** API ingestion, ETL/ELT, batch pipelines, orchestration, data validation, CI/CD, warehouse loading
- **Processing & Modeling:** PySpark, Delta Lake, dbt, PostgreSQL, dimensional modeling, star schema
- **AI / RAG:** document ingestion, chunking, embeddings, vector search, retrieval-augmented generation, cited answers
- **BI & Tools:** Power BI, DAX, Docker, Airflow, GitHub Actions, Git/GitHub, Jupyter Notebook, VS Code
- **AI Coding Tools:** Codex, Claude Code, GitHub Copilot

Projects

1. NYC 311 Service Requests Lakehouse | Azure • Databricks • PySpark • SQL • Delta Lake

Mar 2026 – May 2026 [GitHub](#) | [Case Study](#)

- Built an Azure lakehouse pipeline for NYC 311 analytics using Azure Data Factory, ADLS Gen2, Databricks, Python/PySpark, SQL, and Delta Lake to process Bronze, Silver, and Gold layers.
- Developed reusable ingestion, cleaning, SQL validation, and dimensional modeling logic to produce analytics-ready fact tables, Gold marts, runbooks, architecture notes, and cloud execution proof.

2. Cloud Flight Fare Pipeline | AWS • Python • SQL • ECS/Fargate • S3 • Redshift • dbt

Feb 2026 – May 2026 [GitHub](#) | [Case Study](#)

- Built an end-to-end AWS batch data pipeline using Python, Docker, ECS/Fargate, EventBridge Scheduler, S3, Redshift Serverless, SQL, and dbt to ingest and transform external flight fare API data.
- Developed dbt SQL staging models, dimensional marts, data quality tests, CI checks, architecture diagrams, runbooks, CloudWatch proof logs, and cost/secret safety documentation.

3. CivicLens RAG Operations Copilot | Python • PostgreSQL/pgvector • Streamlit • Docker • RAG

May 2026 [GitHub](#) | [Case Study](#)

- Built a local hybrid RAG copilot that ingests curated NYC 311 documentation, chunks source text, stores embeddings in PostgreSQL/pgvector, and retrieves cited context for grounded answers.
- Developed a Streamlit UI with cited answers, sample analytics responses, source citations, retrieved chunk previews, Dockerized pgvector setup, pytest coverage, GitHub Actions CI, and an 18-question evaluation set.

4. Financial Complaint Auto-Routing with NLP | Python • scikit-learn • NLP • TF-IDF

Jun 2026 – Present [GitHub](#) | [Case Study](#)

- Built a supervised NLP project to classify CFPB consumer complaint narratives into financial product categories for routing support.
- Prepared cleaned complaint text, target labels, EDA outputs, and documentation for future TF-IDF baseline and DistilBERT model comparison.

Experience

IT Intern — Fulton County | Jan 2024 – Apr 2024

- Standardized IT asset and network inventory records, improving data consistency and audit readiness.
- Reconciled data from multiple systems to support inventory reporting, standardization, and internal documentation.

Floral Assistant — Kroger | Dec 2022 – Aug 2023

- Reconciled inventory, pricing, and shrink records to improve reporting accuracy and support exception tracking.

Education

- **Master of Science (M.S.) in Information Technology, Data Analytics Concentration** — Middle Georgia State University, Expected Dec 2026 | GPA: 4.00
- **Associate of Applied Science (A.A.S.) in Website Design/Development** — Gwinnett Technical College, Dec 2024 | GPA: 3.96
- **Bachelor of Arts (B.A.) in Chinese Ethnic Language and Literature** — Guangxi University for Nationalities, China